A GOOD ELEMENT IS A NONZERODIVISOR AND ITS QUOTIENT IS COHEN-MACAULAY WITH A SHORTER MAXIMAL REGULAR SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof refers to two same-chapter lemmas by label (Nakayama and the permutation of a regular sequence); both are recorded as navigation stubs located at their \label lines, and the arguments consuming them stay inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Nakayama's lemma. Invoked in the counted proof at line 25218. Turns the equality f_1 K = K for a finite-length kernel K into K = 0.
- Permuting an M-regular sequence. Invoked in the counted proof at line 25228. Upgrades the regularity on the quotient to regularity of the pair g, f_1 on M.

Source reference. Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 25218); the argument that consumes it is retained in the span.

Source reference. Original lemma 00LJ: sources/stacks-algebra/pinned-original-source.tex, source lines 16856–16856. Navigation pointer for \ref{lemma-permute-xi} used inside the counted span (line 25228); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Notation and assumptions as above. If g is good with respect to $(M, f_1, \ldots, f_d)$, then (a) g is a nonzerodivisor on M, and (b) M/gM is Cohen-Macaulay with maximal regular sequence $f_1, \ldots, f_{d-1}$.*

Proof. We prove the lemma by induction on d. If $d = 0$, then M is finite and there is no case to which the lemma applies. If $d = 1$, then we have to show that $g : M \to M$ is injective. The kernel K has support $\{\mathfrak{m}\}$ because by assumption $\dim \mathrm{Supp}(M) \cap V(g) = 0$. Hence K has finite length. Hence $f_1 : K \to K$ injective implies the length of the image is the length of K, and hence $f_1 K = K$, which by Nakayama's Lemma 00DV implies $K = 0$. Also, $\dim \mathrm{Supp}(M/gM) = 0$ and so M/gM is Cohen-Macaulay of depth 0.

Assume $d > 1$. Observe that g is good for $(M/f_1 M, f_2, \ldots, f_d)$, as is easily seen from the definition. By induction, we have that (a) g is a nonzerodivisor on $M/f_1 M$ and (b) $M/(g, f_1)M$ is Cohen-Macaulay with maximal regular sequence $f_2, \ldots, f_{d-1}$. By Lemma 00LJ we see that g, f_1 is an M-regular sequence. Hence g is a nonzerodivisor on M and $f_1, \ldots, f_{d-1}$ is an M/gM-regular sequence. □